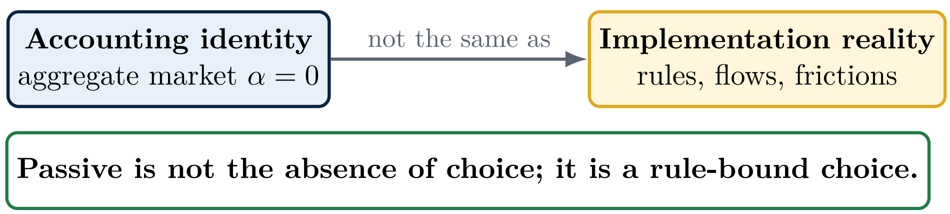
Active vs. Passive Investing

Sharpe's arithmetic is right. The label is not.

OVVO Labs Research Note



Core claim. "Passive" investing is usually described as the neutral alternative to active management. That framing is too generous. A passive portfolio may be low-discretion, low-turnover, and low-cost, but it still embeds active choices about the benchmark, weighting rule, reconstitution schedule, treatment of flows, tax management, securities lending, and tolerance for concentration and momentum exposure. And because the market portfolio itself is a moving target, even the purest indexer must trade.

Revised and extended version of Appendix A of Viole and Nawrocki, *Physics Envy* (2011). The argument of the original appendix is retained; the statement of the arithmetic has been formalized, the arithmetic of a changing market (Pedersen, 2018) and the price-discovery externality (Grossman and Stiglitz, 1980) have been added, and the empirical record has been brought forward to the mid-2020s.

Executive takeaway

Sharpe's arithmetic of active management is an accounting result: before costs, the average actively managed dollar cannot beat the average passively managed dollar; after costs, the average actively managed dollar must lag. As an aggregate identity under its stated assumptions, it is difficult to dispute.

The problem is not the arithmetic. The problem is the label. The phrase "passive investor" does more work than the mathematics can support. It smuggles in the idea that one class of investors is neutral, inactive, or somehow outside the causal machinery of markets.

Thesis. Sharpe's accounting identity establishes that the aggregate market cannot create excess return against itself. It does not establish that passive investing is decisionless, frictionless, or economically inert. When the identity's assumptions are written down explicitly, each one turns out to conceal a decision that someone must make.

When Appendix A of *Physics Envy* first made this argument in 2011, it was a minority position. The subsequent literature has largely caught up. Pedersen (2018) showed that the identity's hidden assumption of a static market portfolio fails in practice: index membership, share issuance, and buybacks force even the purest indexer to trade, and active managers can earn fair compensation supplying liquidity at exactly those moments. Meanwhile, passive vehicles grew from a large minority in 2011 to the majority of United States fund assets, which moved the Grossman–Stiglitz question — if prices are set by active traders, who sets them when the active sector shrinks? — from a seminar-room curiosity to a live structural issue.

The proper contrast is not active versus passive. The proper contrast is **discretionary choice versus rule-bound choice**. Indexing is often a useful rule-bound choice. But a rule is still a choice, and when that rule is market-cap weighting it systematically accepts concentration, price momentum, sector crowding, and valuation drift as portfolio inputs.

1. The arithmetic is an accounting identity

Sharpe's (1991) classic argument can be summarized in two lines:

> *If active and passive management styles are defined sensibly, then before costs the return on the average actively managed dollar equals the return on the average passively managed dollar; after costs, the average actively managed dollar must be lower.*

The market is the aggregation of its owners. Let w_m denote the market portfolio's weight vector, w_i investor i's portfolio, and a_i investor i's share of total investable wealth. Because every share is held by someone,

$$\sum_i a_i w_i = w_m. \tag{1}$$

Subtracting w_m from both sides, active positions are in zero net supply:

$$\sum_i a_i \left(w_i - w_m\right) = 0. \tag{2}$$

Hence, before costs, the wealth-weighted average return of all investors equals the market return r_m, and the wealth-weighted average *active* return is exactly zero. If active implementation costs c_A exceed passive implementation costs c_P, then after costs

$$\bar{r}_A^{\text{net}} \; = \; r_m - c_A \; < \; r_m - c_P \; = \; \bar{r}_P^{\text{net}}. \tag{3}$$

That is true as far as it goes. But it is an aggregation statement, not an ontology of investor types, and it rests on three assumptions that are rarely stated:

A1. Exhaustive classification. Every dollar is either "passive" (holds exactly w_m) or "active" (holds anything else), and both are measured against the *same* market aggregate. In practice, funds are benchmarked to overlapping sub-indices, and the identity holds only at the level of the total aggregate, not within any benchmark segment.

A2. Static market. The composition of w_m changes only through relative price moves, never through share issuance, repurchases, listings, delistings, or index reconstitution. Section 4 shows this assumption fails, and by an economically meaningful margin.

A3. Dollar-weighted averaging. The claim concerns the wealth-weighted *average* dollar. It is silent about the distribution around that average, and therefore silent about whether identifiable subsets of investors possess skill (Section 6).

Key distinction. "The market has zero alpha against itself" is not the same statement as "some investors make no active choices." Equation (2) constrains the wealth-weighted sum of deviations. It says nothing about how any individual portfolio came to be.

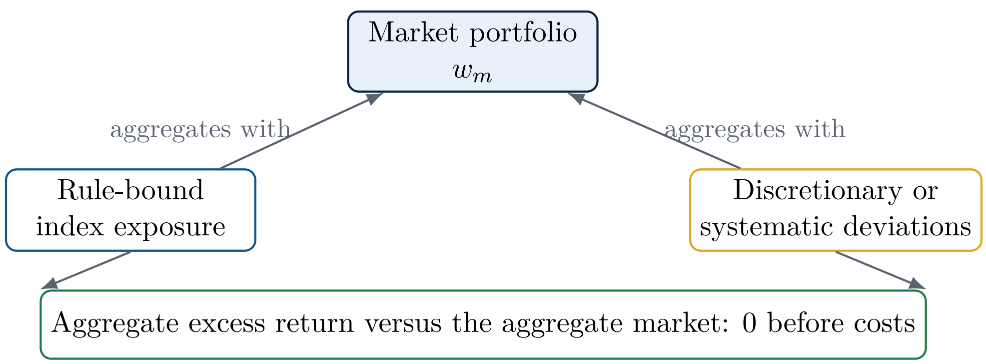


Figure 1: Sharpe's arithmetic is an aggregate accounting identity. It says that active weights net to zero before costs. It does not prove that one side of the implementation spectrum is choice-free.

2. The classification problem

The usual definition of a passive investor is approximately this: a passive investor holds every security in the market in the same proportion as the market. If security X is 3 percent of total market value, then the passive investor holds 3 percent of the portfolio in X.

As a limiting definition, this is elegant. As a realized investor classification, it is too clean.

A real portfolio has:

- a chosen benchmark,
- a chosen weighting rule,
- a chosen reconstitution calendar,
- a method for handling inflows and outflows,

- a policy for taxes, dividends, corporate actions, and securities lending,
- a tolerance for tracking error, sampling error, and execution costs.

Each item is a decision. Some decisions are automated, some are outsourced, and some are embedded in index methodology. But automation does not erase agency. It merely converts discretion into rules.

Better definition. Passive investing is not non-action. It is low-discretion rule-following, usually benchmarked to a capitalization-weighted index and implemented with the objective of minimizing cost and tracking error.

This is why the active/passive binary is misleading. Most investors sit on a spectrum. A capitalization-weighted index fund, an optimized sampling ETF, a factor index, a tax-managed direct indexing account, a trend-following overlay, and a discretionary stock picker differ by the form and amount of discretion. They are not separated by the presence or absence of choice.

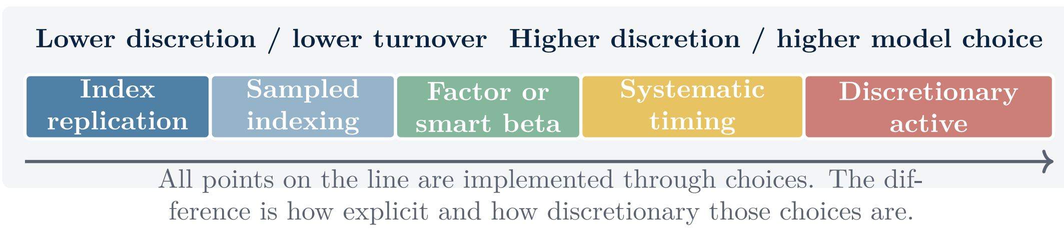


Figure 2: The economically useful dimension is not active versus passive. It is the amount and type of discretion embedded in the rule set.

Assumption A1 fails along this spectrum in a second way as well. Most "active" funds are not benchmarked to the total market but to style boxes, sectors, regions, or custom blends. Sharpe's identity binds each properly defined aggregate against itself; it does not adjudicate the contest between a mid-cap value manager and the total market, because those are different aggregates. Much of the popular active-versus-passive scorekeeping quietly changes the benchmark mid-argument.

3. A technical correction: cap-weighted portfolios do not rebalance against price appreciation

A common criticism of passive investing says that cap-weighted funds are forced to buy what has already gone up and sell what has already gone down. That criticism is directionally intuitive, but it must be stated carefully, because as stated it is false.

The algebra makes it unambiguous. Suppose a fund holds the same fixed fraction θ of every firm's shares outstanding: θs_j shares of firm j, where s_j is the firm's share count. Then the fund's weight in firm j is

$$w_j = \frac{p_j\,(\theta s_j)}{\sum_k p_k\,(\theta s_k)} = \frac{p_j s_j}{\sum_k p_k s_k}, \tag{4}$$

which is the market weight identically, for *any* vector of prices p. When a price moves, the weight moves with it, automatically, with zero shares traded. A true cap-weighted portfolio is self-rebalancing against price changes.

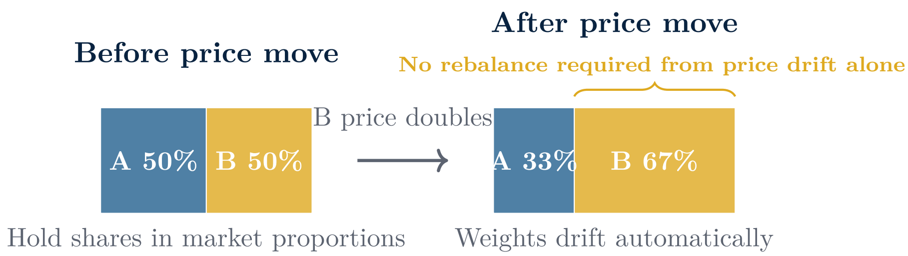


Figure 3: A true cap-weighted portfolio absorbs relative price moves through mark-to-market changes. If B's price doubles, its weight rises from 50 to 67 percent with no trading at all. The portfolio is not forced to buy the winner.

The stronger criticism is different:

Stronger criticism. A market-cap-weighted rule mechanically accepts the market's prior price action as the new portfolio exposure. It does not rebalance against price movement; it institutionalizes price movement into weights.

Equation (4) also identifies exactly when the indexer *must* trade: whenever s_j changes (issuance, buybacks) or the set of firms changes (listings, delistings, index reconstitution) — and whenever the fund's own θ changes through investor inflows and outflows. Price changes never force the indexer to trade; share-count and membership changes always do. That observation is the bridge to the next section.

4. The market portfolio is a moving target

Assumption A2 — the static market — is where Sharpe's arithmetic meets its most consequential post-2011 refinement. Pedersen (2018) pointed out that the market portfolio is not a fixed object: firms issue and repurchase shares, list and delist, enter and exit indices. An investor who wants to keep holding "the market" must therefore trade, and not trivially. Pedersen estimates that simply maintaining a value-weighted portfolio of United States equities has historically required turnover of roughly 7.6 percent per year.

His thought experiment makes the point vivid. An investor who bought the entire United States equity market in 1926 and never traded again would today hold only the dwindling subset of firms that have survived since 1926 — a museum piece, not the market. Buy-and-hold and market-tracking are different strategies, and only trading reconciles them.

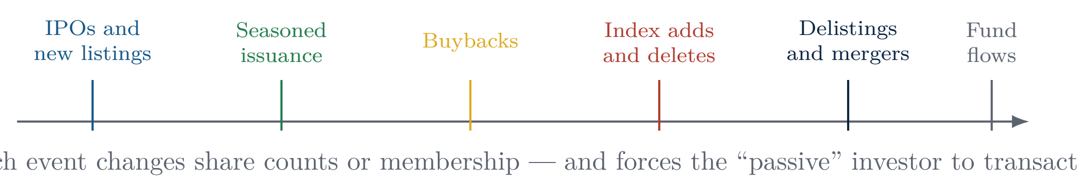


Figure 4: The market portfolio's composition changes continuously. Pedersen (2018): tracking it has historically required turnover on the order of 7.6 percent of United States equity value per year.

Three consequences follow.

1. **Pure passivity is unattainable.** Every index fund transacts at reconstitutions, corporate actions, and flows, and every transaction has a counterparty, a price, and a cost. The frictionless passive dollar of equation (3) is a limit, not a portfolio anyone holds.

2. **The zero-sum conclusion softens.** When passive investors must trade, informed active managers can earn genuine compensation — not by winning a zero-sum bet against other traders, but by supplying liquidity and price discovery at issuance, at buybacks, and at index events. Active management in aggregate can add value up to (and in equilibrium, approximately equal to) the cost of providing those services.

3. **Timing choices re-enter through the back door.** *When* an index reconstitutes, *how* an index fund trades into the reconstitution, and *whether* it front-runs, matches, or lags the official close are implementation decisions with measurable return consequences. The rule does not remove the decisions; it assigns them to the index committee and the trading desk.

The unifying algebra. Sections 3 and 4 are two sides of equation (4). Price changes never force the indexer to trade — so the naive "passive buys winners" critique fails. Share-count and membership changes always force the indexer to trade — so the naive "passive never trades" defense fails too.

5. The missing reference point

A further hidden assumption in clean active/passive arithmetic is that the passive portfolio can be defined as if it exists outside transaction time. In reality, each investor has a reference point: the date and price at which capital is committed, withdrawn, inherited, taxed, rebalanced, or benchmarked.

A market portfolio may be an aggregate object. An investor's portfolio is a path-dependent object.

Every implementation creates reference points. Reference points create realized investor-specific returns.

Figure 5: The aggregate market is not path-dependent against itself, but each realized investor account is path-dependent through contributions, withdrawals, tax events, and implementation dates.

This is where the binary classification weakens. If a so-called passive fund receives inflows and must put that capital to work, it is transacting. If it tracks an index whose composition changes, it is transacting. If it samples rather than fully replicates, it is optimizing. If it lends securities, manages tax lots, or chooses a benchmark, it is making economic choices.

The choices may be mechanical and sensible. They may even be superior to discretionary active management for many investors. But they are still choices.

6. Who sets the prices? The Grossman–Stiglitz limit

The indexing rule contains a quiet dependency: it takes market prices as given and correct enough to weight by. But prices do not set themselves. Grossman and Stiglitz (1980) showed that perfectly efficient prices are impossible in equilibrium: if prices fully reflected all information, no one would pay to gather information, and then prices could not reflect it. Some level of compensated active management is required for the price system the indexer free-rides on.

The free-rider framing. Price discovery is a positive externality produced by the active sector and consumed, free of charge, by the passive sector. Indexing is a brilliant piece of institutional engineering precisely *because* it free-rides. But a strategy that depends on the existence of the thing it displaces cannot be scaled to 100 percent of the market.

This does not tell us where the equilibrium sits, and the honest answer is that nobody knows. Two empirical points from the years since the original appendix discipline the debate in both directions:

- **Passive is no longer the minority.** Assets in United States index equity funds overtook their actively managed counterparts around 2019, and by 2024 passive assets exceeded active across United States mutual funds and ETFs as a whole. The Grossman–Stiglitz question is no longer hypothetical. Relatedly, the combined index weight of the largest handful of United States stocks reached multi-decade highs in the mid-2020s — concentration a cap-weighted rule accepts by construction.
- **But do not overclaim the distortion.** The classic "index effect" — the price pop when a stock joins a major index — has *declined* toward zero over the same decades that indexing grew (Greenwood and Sammon, 2025). Markets appear to have arbitraged the most mechanical consequence of index flows even as those flows grew. Critics of indexing who predict ever-larger mechanical distortions must contend with this evidence.

Assumption A3 deserves its own footnote here. The identity constrains the average dollar, not every dollar. Berk and Green (2004) showed that genuine manager skill is fully consistent with zero average net alpha for investors: flows chase skill until fees and scale absorb it, so the rents accrue to the skilled manager rather than the fund investor. Fama and French (2010) found that in aggregate, United States equity mutual funds deliver net-of-fee alpha close to the negative of their costs, with only weak evidence of skill beyond luck in the extreme tails — exactly what equations (2) and (3) plus Berk–Green competition would predict. And French (2008) priced the whole contest: society's expenditure on active price discovery ran to roughly 0.67 percent of aggregate United States market capitalization per year. Zero average alpha is not evidence that nothing is happening; it is evidence that a large, costly, competitive industry is doing something the arithmetic nets to zero.

7. What passive really means

The cleanest description is this:

Passive investing is indeed active, albeit infrequently. It is a constrained active rule: active in design, and passive only in day-to-day discretion.

The investment committee, index provider, fund sponsor, and investor make decisions before the rule runs. Once the rule is running, the portfolio behaves mechanically. That is not absence of activity; it is delegated activity.

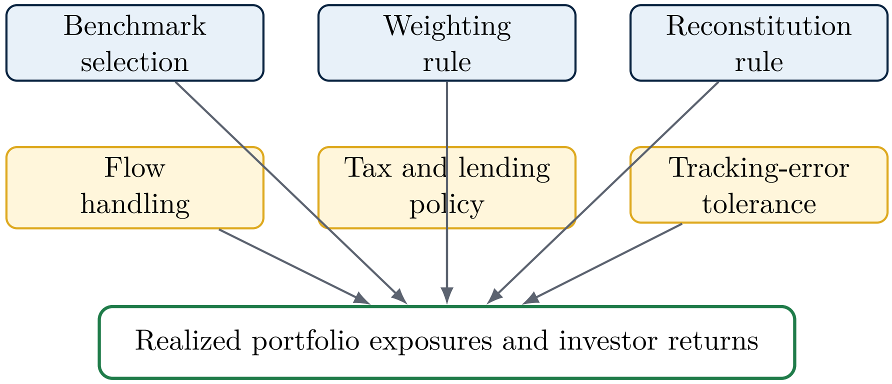


Figure 6: The passive outcome is the output of several active design choices. The discretion is front-loaded into the rule set.

This framing does not deny the empirical advantages of low-cost indexing. It clarifies them. Indexing often wins because it reduces costs, taxes, turnover, manager-selection error, and behavioral mistakes. Those are real advantages. But those advantages do not require pretending that passive investors are metaphysically passive.

8. Replacing the old color analogy

The original appendix leaned on a color analogy: combine passive investors and active investors and the total market becomes a single aggregate color with zero alpha. That is fine as an accounting picture, but it is misleading as a classification picture, because it implies two primary colors when the reality is a continuous blend.

A better picture is a spectrum of rule sets. Some investors hold broad cap-weighted portfolios. Some hold optimized index samples. Some hold factor screens. Some use systematic timing. Some use discretionary views. The aggregate market is the wealth-weighted result of all of them.

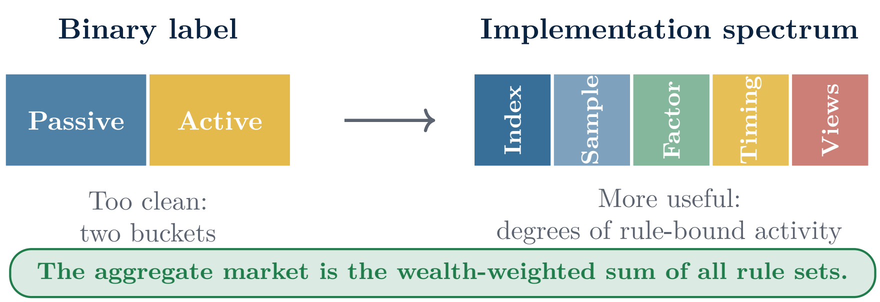


Figure 7: The active/passive binary hides the fact that all realized portfolios are implemented through choices. The useful distinction is the intensity and form of the rule.

9. Why this distinction matters

The passive label matters because labels shape investor behavior. Once a portfolio is called passive, investors can mistake it for neutral. But a capitalization-weighted portfolio is not neutral with respect to concentration, valuation, sector exposure, or momentum. It is explicitly designed to own more of what has become larger by market value.

What the cap-weighted rule buys you	What it exposes you to
Low turnover under ordinary price movement	Increasing concentration when a few securities dominate market capitalization
Broad diversification	Pro-cyclical exposure to prior winners
Transparent construction	Embedded sector drift
Low fees	Benchmark-provider dependency
Limited manager discretion	Sensitivity to flows and index events
High tax efficiency in many implementations	No intrinsic valuation discipline

None of these points refutes Sharpe. They refine him. The aggregate market remains the aggregate market. But a real investor does not own an abstraction. A real investor owns a path-dependent implementation of a chosen rule.

10. A better version of the conclusion

The standard conclusion is:

> *Because active management is zero-sum before costs and negative-sum after costs, passive management is the rational default.*

A stronger and more precise conclusion is:

> *Because aggregate active weights net to zero before costs and lose to costs after costs, investors should be skeptical of expensive discretionary deviation. But low-cost index exposure is still a chosen rule set, not the absence of investment choice; the market it tracks is a moving target that forces even indexers to trade; and the prices it weights by are produced by the active sector it free-rides on. The relevant question is whether the chosen rule produces the exposures the investor actually wants, at a cost the exposures are worth.*

That version keeps the accounting result while removing the mythology.

Final formulation. Passive investing is not free from active assumptions. It is an active decision to accept the market's capitalization-weighted assumptions at low cost — a decision that has usually been a good one, for reasons the arithmetic explains and the label obscures.

One-page summary

Claim	Clean version
Sharpe's arithmetic	Correct as aggregate accounting: the market cannot create excess return against itself before costs.
Hidden assumptions	Exhaustive two-way classification (A1), a static market portfolio (A2), and dollar-weighted averaging (A3). Each conceals a decision or an escape hatch.
Weakness in the label	The arithmetic does not prove that passive investors make no active choices.
Best definition	Passive investing is indeed active, albeit infrequently: low-discretion, rule-bound implementation, usually designed to minimize cost and tracking error.
Technical correction	Cap-weighted portfolios do not rebalance because a constituent appreciates; the weight changes automatically with zero trading.
Moving-target result	Share issuance, buybacks, and reconstitution force even pure indexers to trade (roughly 7.6% annual turnover historically; Pedersen 2018), so active management can earn fair compensation supplying that liquidity.
Price-discovery limit	Indexing free-rides on prices set by the active sector (Grossman–Stiglitz 1980); it cannot scale to the whole market.
Stronger critique	Cap-weighting institutionalizes prior price action into future portfolio exposure.
Investor implication	The real question is not "active or passive?" It is "which rule, which benchmark, which exposures, and at what cost?"

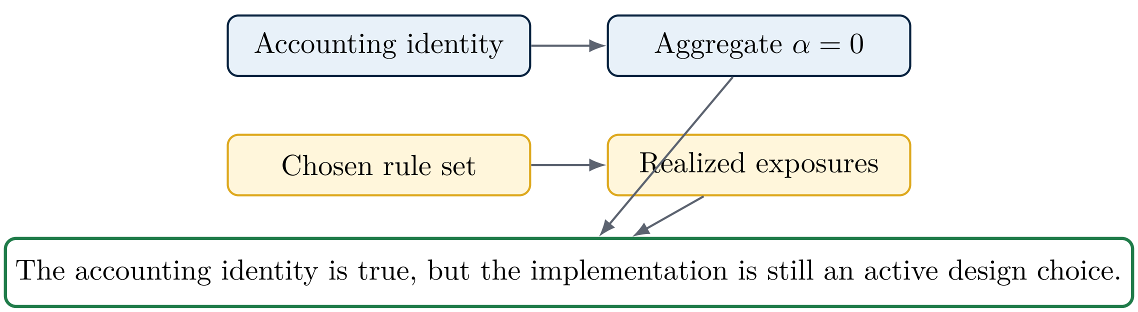


Figure 8: The note in one picture: aggregate arithmetic and portfolio implementation are different categories.

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